TP2 :

Mission 1.1 :

Configuration mot de passe user :

```
[root@srv-core ~]# cp -a /etc/login.defs /root/login.defs.orig
[root@srv-core ~]# sed -i \
-e 's/^PASS_MAX_DAYS.*/PASS_MAX_DAYS   90/' \
-e 's/^PASS_MIN_DAYS.*/PASS_MIN_DAYS   1/' \
-e 's/^PASS_WARN_AGE.*/PASS_WARN_AGE   14/' \
-e 's/^UMASK.*/UMASK           027/' \
/etc/login.defs
[root@srv-core ~]# grep -E '^(PASS_|UMASK|USERGROUPS_ENAB|CREATE_HOME)' /etc
/login.defs
UMASK           027
PASS_MAX_DAYS   90
PASS_MIN_DAYS   1
PASS_WARN_AGE   14
USERGROUPS_ENAB yes
CREATE_HOME     yes
```

```
[root@srv-core ~]# cat > /etc/security/pwquality.conf.d/50-helios.conf <<'EO
F'
> minlen = 12
minclass = 3
maxrepeat = 3
dictcheck = 1
usercheck = 1
enforce_for_root
EOF
[root@srv-core ~]# cat /etc/security/pwquality.conf.d/50-helios.conf
minlen = 12
minclass = 3
maxrepeat = 3
dictcheck = 1
usercheck = 1
enforce_for_root
[root@srv-core ~]#
```

Mission 1.2 :

Création de Groupes, Utilisateurs et mot de passe :

```
[root@srv-core ~]# chage -l julien
Last password change                                    : nov. 03, 2026
Password expires                                        : févr. 01, 2027
Password inactive                                       : févr. 08, 2027
Account expires                                         : never
Minimum number of days between password change          : 1
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 14
[root@srv-core ~]# getent passwd julien nadia tarek audit-ext
julien:x:5100:5000:Julien Marty - chef equipe transport:/home/julien:/bin/ba
sh
nadia:x:5101:5000:Nadia Belkacem - exploitante:/home/nadia:/bin/bash
tarek:x:5102:5001:Tarek Oualid - analyste telemetrie:/home/tarek:/bin/bash
audit-ext:x:5103:5001:Auditeur externe - mandat Q4:/home/audit-ext:/bin/bash
[root@srv-core ~]# getent group transport telemetry helios-ops
transport:x:5000:tarek
telemetry:x:5001:
helios-ops:x:5002:julien
[root@srv-core ~]#
```

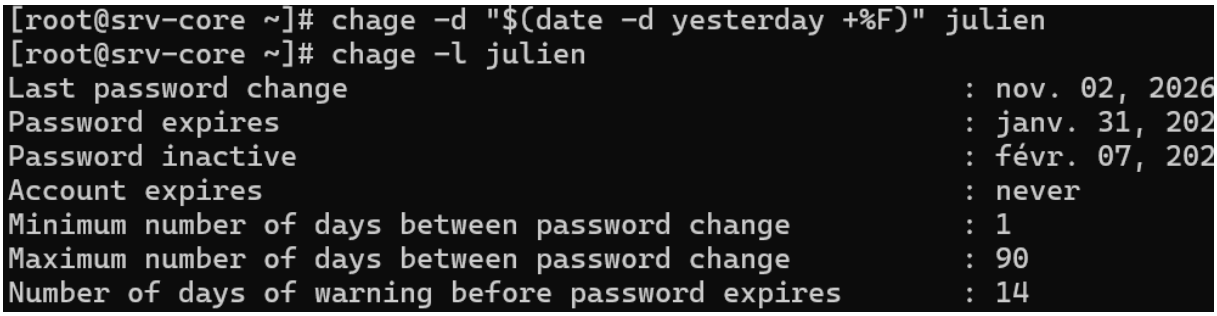
Mission 1.3 :

Julien devient l'admin du groupe Transport :

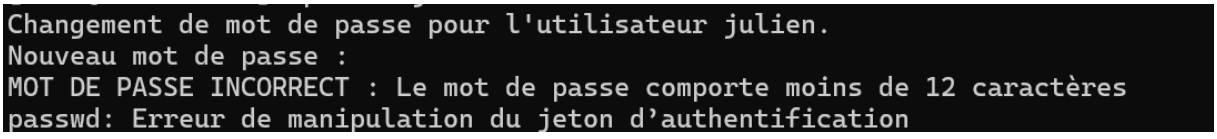


Ajout de Nadia dans le groupe Transport :



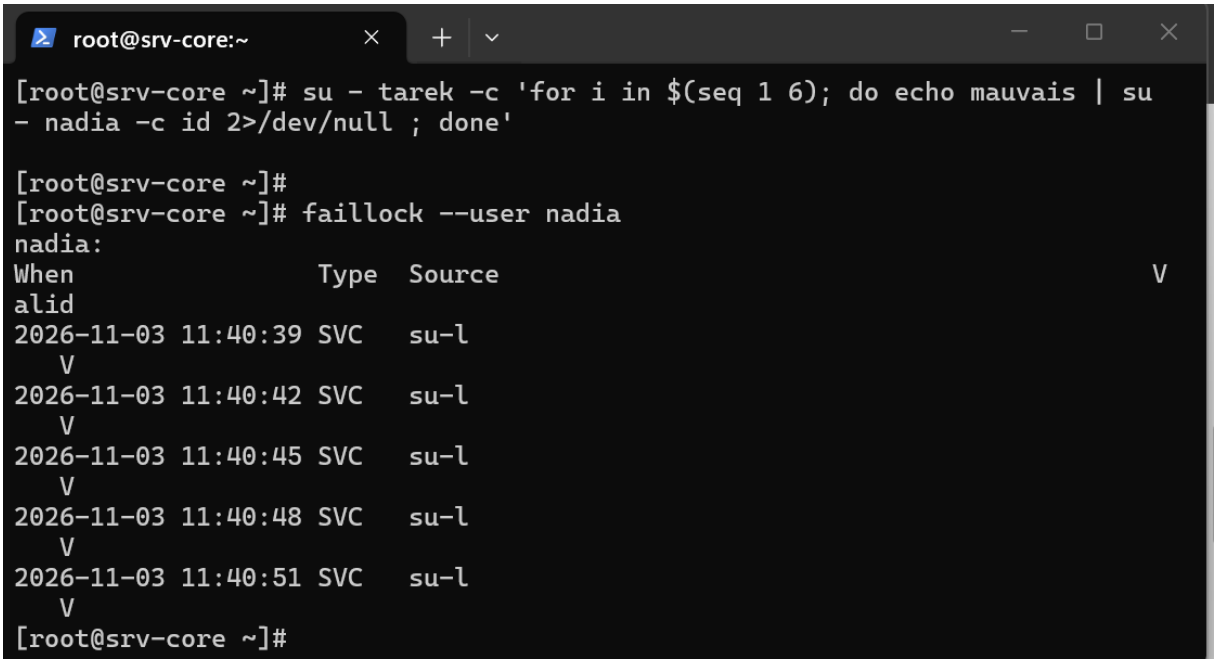


Changement de mot de passe pour Julien, mais mot de passe pas assez fort :

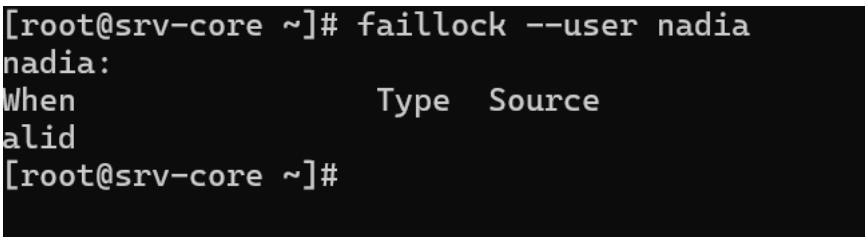


Faillock :

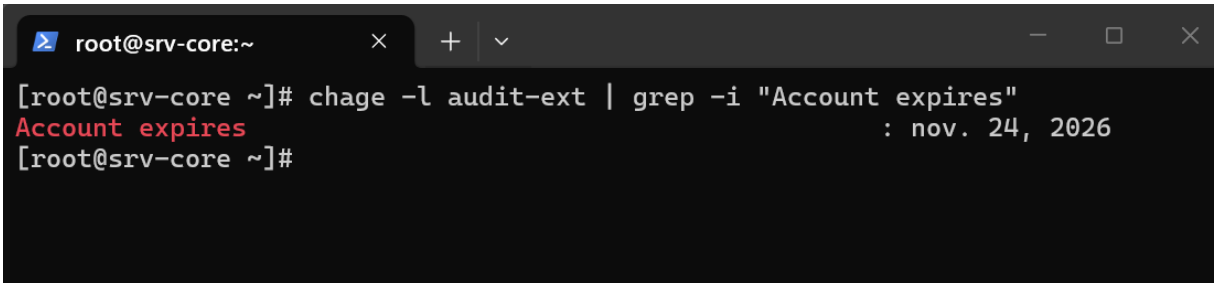
Avant :



Après :

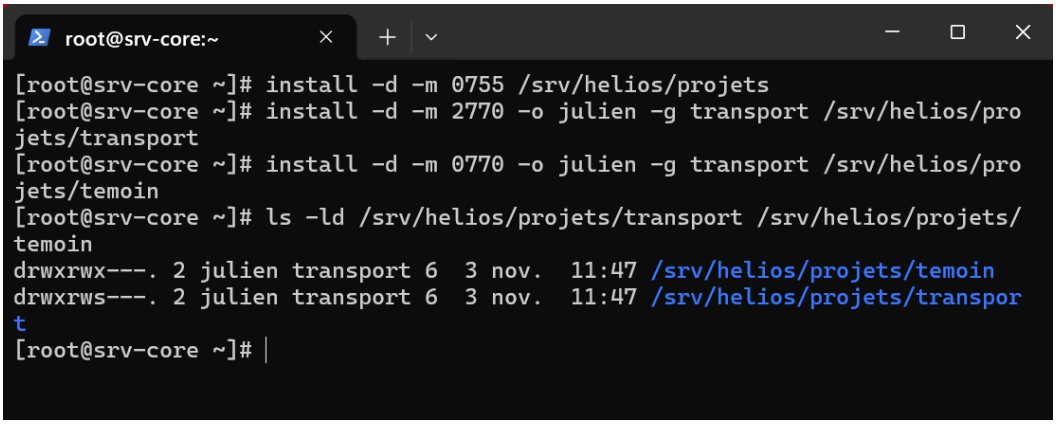


Date d'expiration de l'audit :

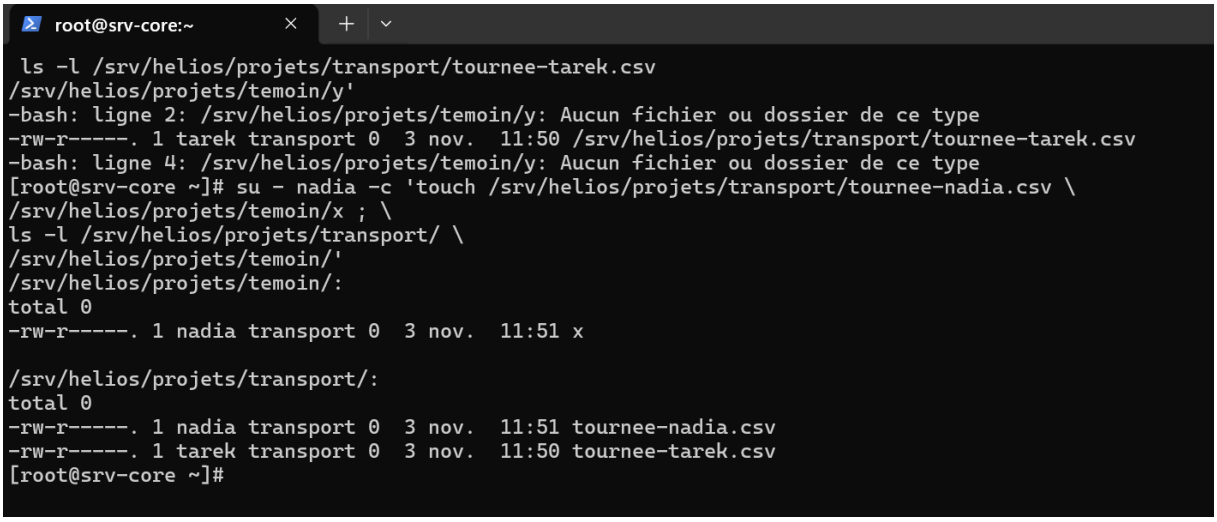


Mission 2.1 :

Création des répertoires :



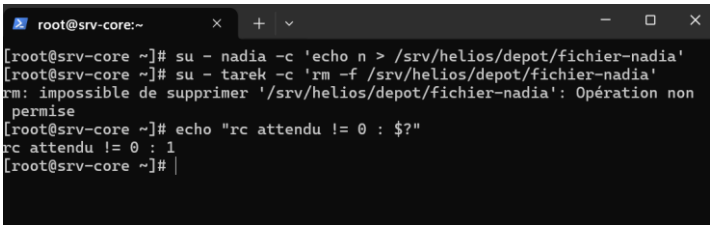
Démonstration SETGID :



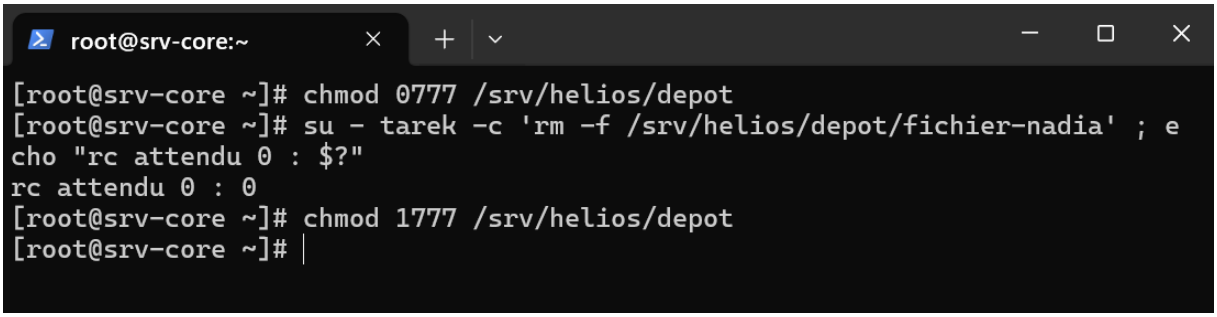
Preuve Sticky Bit :



Refus de suppression Tarek – Nadia :



Démonstration sans Sticky Bit :

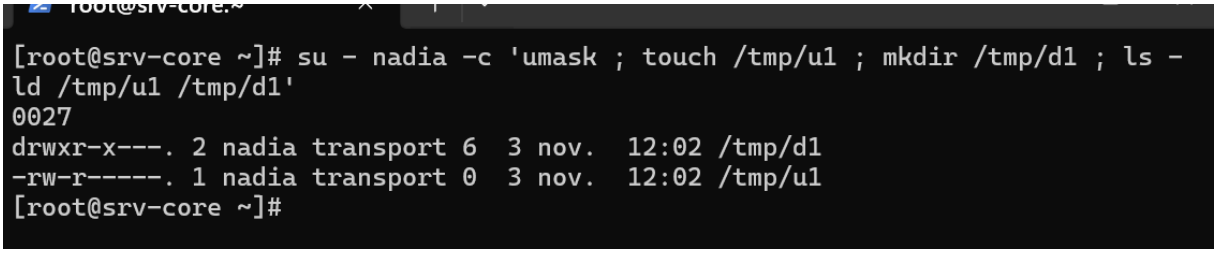


Mission 2.2 :

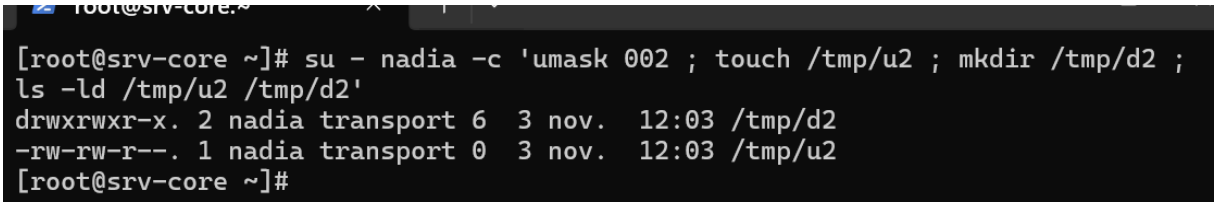
Umask prediction :

Umask	Fichier attendu	Répertoire attendu
022	644	755
002	664	775
077	600	700

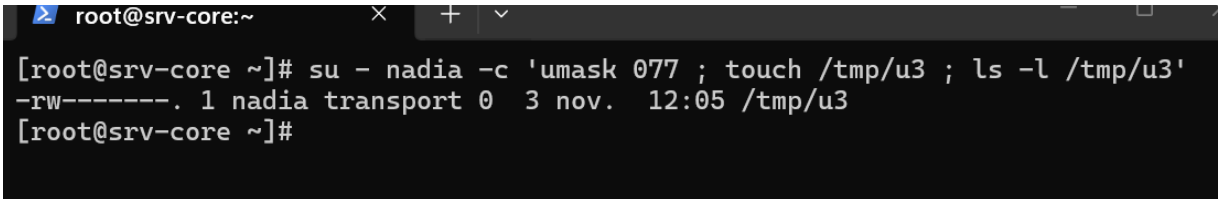
Umask Test 1 Nadia :



Umask Test 2 Nadia :



Umask Test 3 Nadia :



Verif Umask Nadia et audit:

```
[root@srv-core ~]# cat > /etc/profile.d/helios-umask.sh <<'EOF'
> if id -nG | tr ' ' '\n' | grep -qx transport ; then umask 002 ; else umask 0
27 ; fi
> EOF
[root@srv-core ~]# chmod 0644 /etc/profile.d/helios-umask.sh
[root@srv-core ~]# su - nadia -c 'umask'
0002
[root@srv-core ~]# su - nadia -c 'touch /srv/helios/projets/transport/partage.
txt ; ls -l /srv/helios/projets/transport/partage.txt'
-rw-rw-r--. 1 nadia transport 0  3 nov.  12:07 /srv/helios/projets/transport/p
artage.txt
[root@srv-core ~]# su - audit-ext -c 'umask'
0027
[root@srv-core ~]#
```

Mission 2.3 :

Inventaire Setuid :

```
[root@srv-core ~]# find / -xdev \( -perm -4000 -o -perm -2000 \) -type f -prin
tf '%M %u %g %p\n' 2>/dev/null | sort
-rwsr-xr-x root root /usr/bin/chage
-rwsr-xr-x root root /usr/bin/crontab
-rwsr-xr-x root root /usr/bin/gpasswd
-rwsr-xr-x root root /usr/bin/mount
-rwsr-xr-x root root /usr/bin/newgrp
-rwsr-xr-x root root /usr/bin/passwd
-rwsr-xr-x root root /usr/bin/su
-rwsr-xr-x root root /usr/bin/umount
-rwsr-xr-x root root /usr/sbin/grub2-set-bootflag
-rwsr-xr-x root root /usr/sbin/pam_timestamp_check
-rwsr-xr-x root root /usr/sbin/unix_chkpwd
-rwxr-sr-x root tty /usr/bin/write
-rwx--s--x root utmp /usr/libexec/utempter/utempter
-r-xr-sr-x root ssh_keys /usr/libexec/openssh/ssh-keysign
---s--x--x root root /usr/bin/sudo
[root@srv-core ~]#
```

Preuve noyau Linux ignore le bit setuid sur un script interprété :

```
[root@srv-core ~]# printf '#!/usr/bin/bash\nid\n' > /usr/local/bin/faux-setuid
.sh
[root@srv-core ~]# chmod 4755 /usr/local/bin/faux-setuid.sh
[root@srv-core ~]# ls -l /usr/local/bin/faux-setuid.sh
-rwsr-xr-x. 1 root root 19  3 nov.  12:11 /usr/local/bin/faux-setuid.sh
[root@srv-core ~]# su - nadia -c '/usr/local/bin/faux-setuid.sh'
uid=5101(nadia) gid=5000(transport) groupes=5000(transport) contexte=unconfine
d_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[root@srv-core ~]#
```

Les deux sorties de helios-lastlog :

```
[root@srv-core ~]# gcc -O2 -Wall -Wextra \
-o /usr/local/bin/helios-lastlog \
/root/helios-lastlog.c
[root@srv-core ~]# chown root:telemetry /usr/local/bin/helios-lastlog
[root@srv-core ~]# chmod 2755 /usr/local/bin/helios-lastlog
[root@srv-core ~]# ls -l /usr/local/bin/helios-lastlog
-rwxr-sr-x. 1 root telemetry 17504  3 nov.  12:15 /usr/local/bin/helios-lastlo
g
[root@srv-core ~]#
```

```
root@srv-core:~
[root@srv-core ~]# su - julien -c 'tail -n 3 /var/log/helios/app.log' ; echo "
rc attendu != 0 : $?"
tail: impossible d'ouvrir '/var/log/helios/app.log' en lecture: Permission non
 accordée
rc attendu != 0 : 1
[root@srv-core ~]# su - julien -c '/usr/local/bin/helios-lastlog' ; echo "rc a
ttendu 0 : $?"
2026-11-03T12:14:07+01:00 INFO trame 21 ok
2026-11-03T12:14:07+01:00 INFO trame 22 ok
2026-11-03T12:14:07+01:00 INFO trame 23 ok
2026-11-03T12:14:07+01:00 INFO trame 24 ok
2026-11-03T12:14:07+01:00 INFO trame 25 ok
2026-11-03T12:14:07+01:00 INFO trame 26 ok
2026-11-03T12:14:07+01:00 INFO trame 27 ok
2026-11-03T12:14:07+01:00 INFO trame 28 ok
2026-11-03T12:14:07+01:00 INFO trame 29 ok
2026-11-03T12:14:07+01:00 INFO trame 30 ok
rc attendu 0 : 0
[root@srv-core ~]#
```

Mission 3.1 :

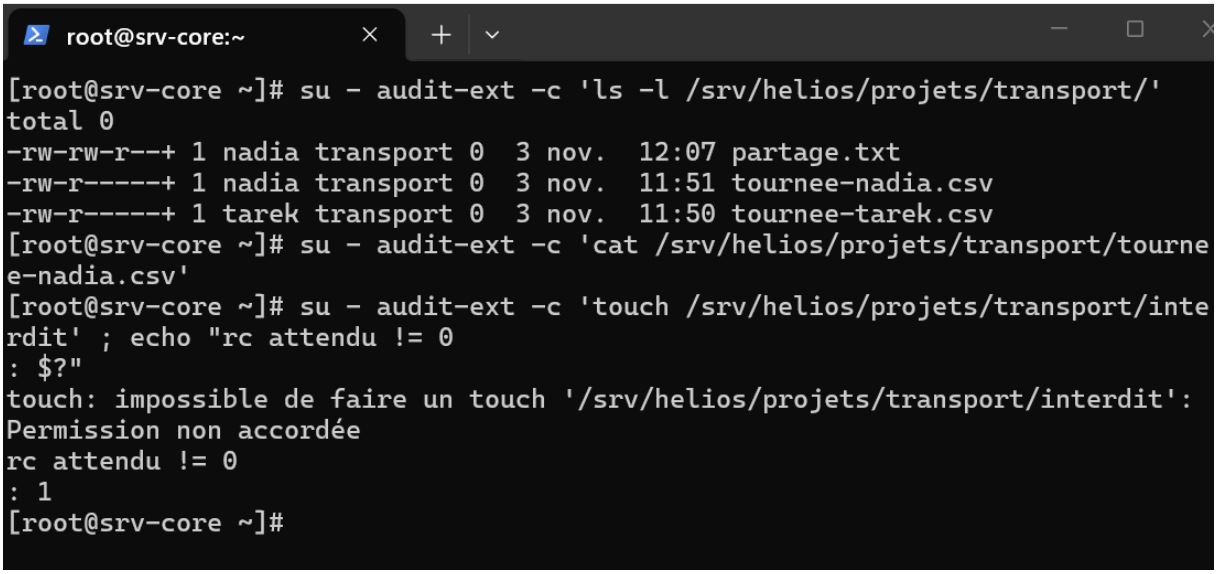
Accès nominatif hors groupe :

```
root@srv-core:~
[root@srv-core ~]# getfacl /srv/helios/projets/transport
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport
# owner: julien
# group: transport
# flags: -s-
user::rwx
group::rwx
other::---

[root@srv-core ~]# setfacl -R -m u:audit-ext:rX /srv/helios/projets/transport
[root@srv-core ~]# getfacl /srv/helios/projets/transport
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport
# owner: julien
# group: transport
# flags: -s-
user::rwx
user:audit-ext:r-x
group::rwx
mask::rwx
other::---

[root@srv-core ~]# ls -ld /srv/helios/projets/transport
drwxrws---+ 2 julien transport 75  3 nov.  12:07 /srv/helios/projets/transport
[root@srv-core ~]#
```

Verfi perm denied :



Mission 3.2 :

Ajouts des ACL :

```
[root@srv-core ~]# setfacl -d -m u:audit-ext:r /srv/helios/projets/transport
[root@srv-core ~]# setfacl -d -m g:transport:rwx /srv/helios/projets/transport
[root@srv-core ~]# setfacl -d -m o::--- /srv/helios/projets/transport
[root@srv-core ~]# getfacl /srv/helios/projets/transport
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport
# owner: julien
# group: transport
# flags: -s-
user::rwx
user:audit-ext:r-x
group::rwx
mask::rwx
other::---
default:user::rwx
default:user:audit-ext:r--
default:group::rwx
default:group:transport:rwx
default:mask::rwx
default:other::---

[root@srv-core ~]#
```

ACL hérité automatiquement :

```
[root@srv-core ~]# su - nadia -c 'touch /srv/helios/projets/transport/tournee-j2.csv ; mkdir -p /srv/helios/projets/transport/semaine-41'
[root@srv-core ~]# getfacl /srv/helios/projets/transport/tournee-j2.csv
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport/tournee-j2.csv
# owner: nadia
# group: transport
user::rw-
user:audit-ext:r--
group::rwx                      #effective:rw-
group:transport:rwx             #effective:rw-
mask::rw-
other::---

[root@srv-core ~]# getfacl /srv/helios/projets/transport/semaine-41
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport/semaine-41
# owner: nadia
# group: transport
# flags: -s-
user::rwx
user:audit-ext:r--
group::rwx
group:transport:rwx
mask::rwx
other::---
default:user::rwx
default:user:audit-ext:r--
default:group::rwx
default:group:transport:rwx
default:mask::rwx
default:other::---

[root@srv-core ~]# su - audit-ext -c 'cat /srv/helios/projets/transport/tournee-j2.csv' ; echo "rc attendu 0
: $?"
rc attendu 0
: 0
[root@srv-core ~]#
```

Mission 3.3 :

Avant :

```
[root@srv-core ~]# setfacl -m u:tarek:rwx /srv/helios/projets/transport/tournee-j2.csv
[root@srv-core ~]# getfacl /srv/helios/projets/transport/tournee-j2.csv
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport/tournee-j2.csv
# owner: nadia
# group: transport
user::rw-
user:tarek:rwx
user:audit-ext:r--
group::rwx
group:transport:rwx
mask::rwx
other::---

[root@srv-core ~]# chmod g-w /srv/helios/projets/transport/tournee-j2.csv
[root@srv-core ~]# getfacl /srv/helios/projets/transport/tournee-j2.csv
getfacl : suppression du premier « / » des noms de chemins absolus
# file: srv/helios/projets/transport/tournee-j2.csv
# owner: nadia
# group: transport
user::rw-
user:tarek:rwx                  #effective:r-x
user:audit-ext:r--
group::rwx                      #effective:r-x
group:transport:rwx             #effective:r-x
mask::r-x
other::---

[root@srv-core ~]#
```

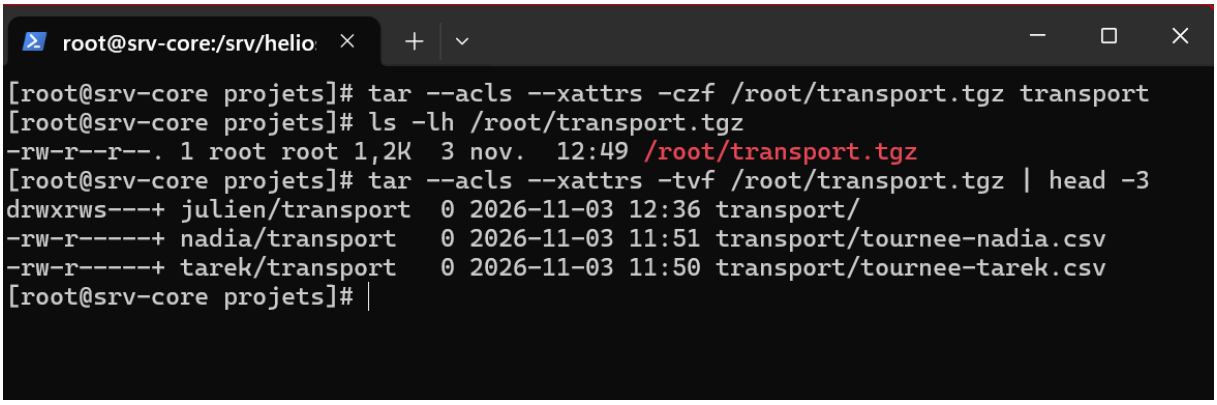
Après :



Chmod g-w a modifié le Mask ACL et non l'entrée ACL de Tarek. L'ACL indique toujours rwx, mais le masque retire l'écriture, ce qui rend les droits effectifs r-x. Tarek perd donc l'autorisation d'écrire jusqu'à ce que le masque soit remis à rwx.

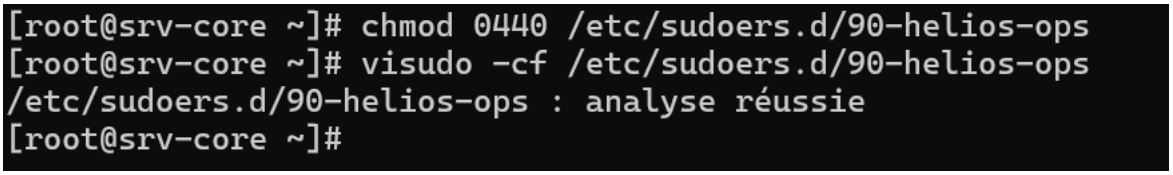
Mission 3.4 :

ACL :



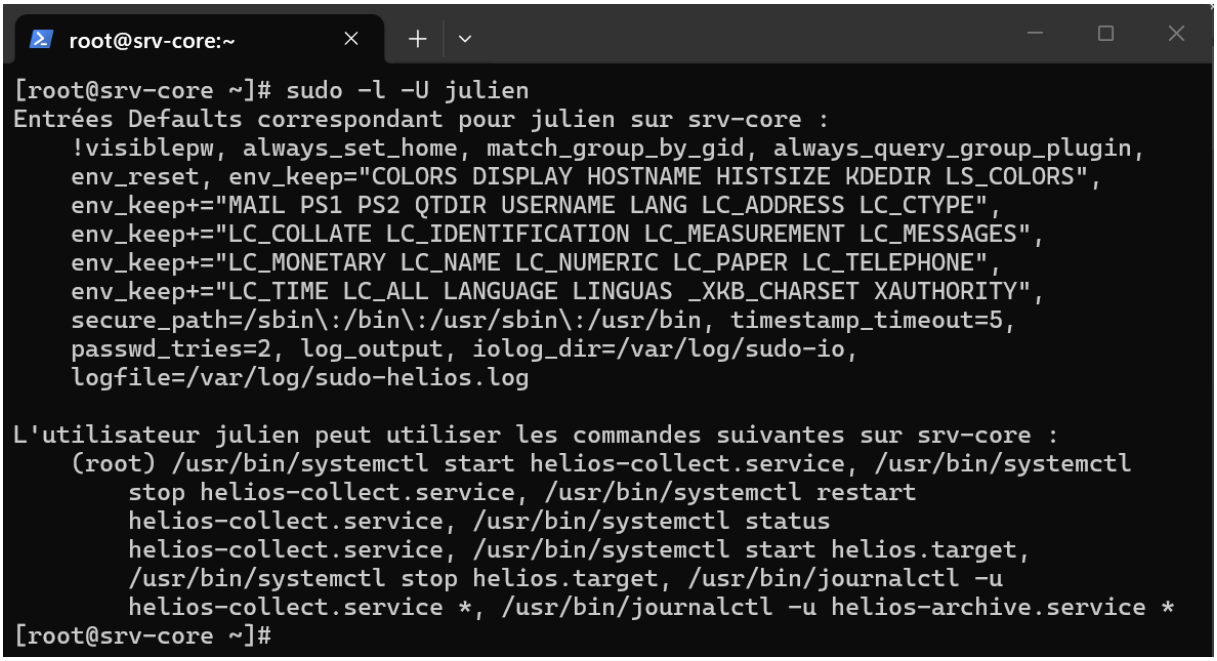
Mission 4.1 :

90-helios-ops validé par visudo :

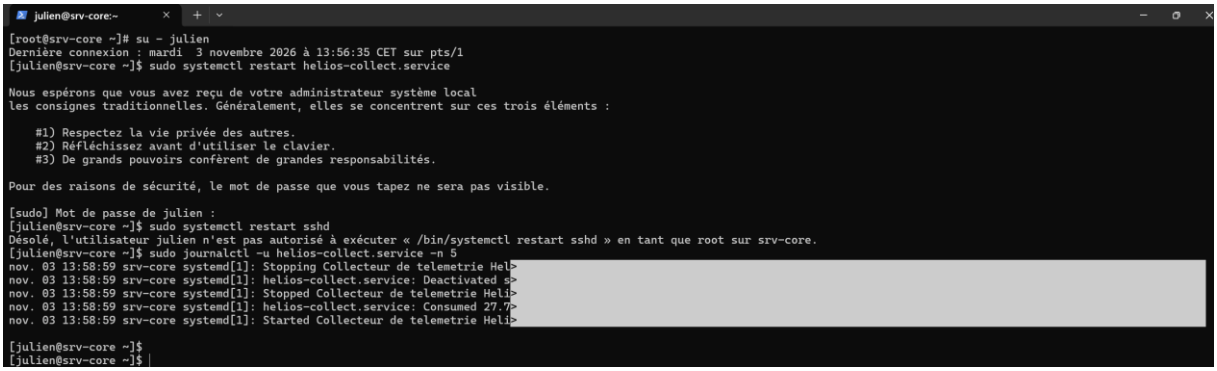


Mission 4.2 :

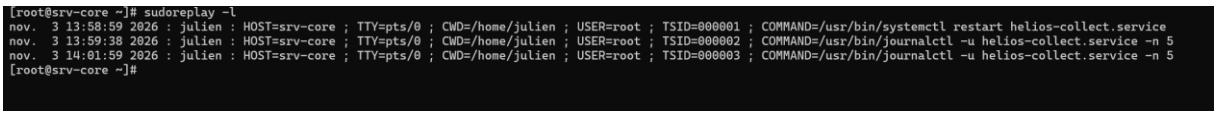
Droits de julien :



Verif droits de julien :

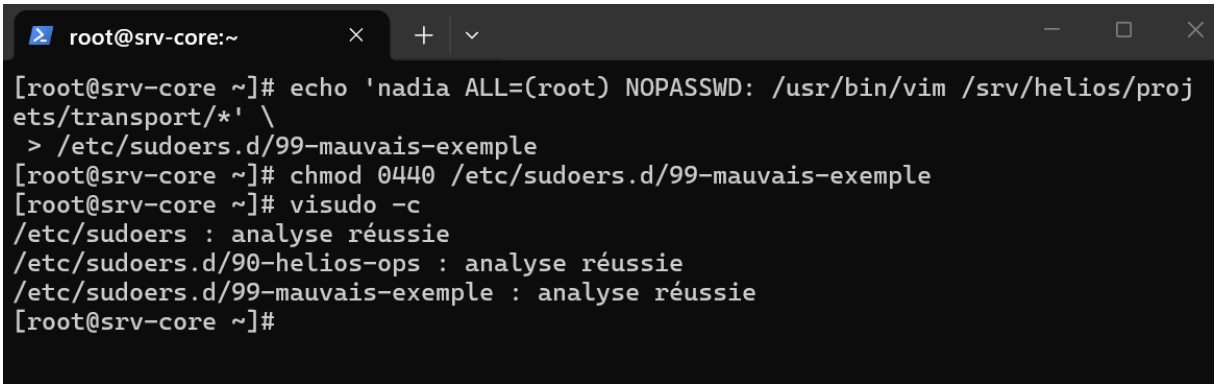


Sudoreplay :

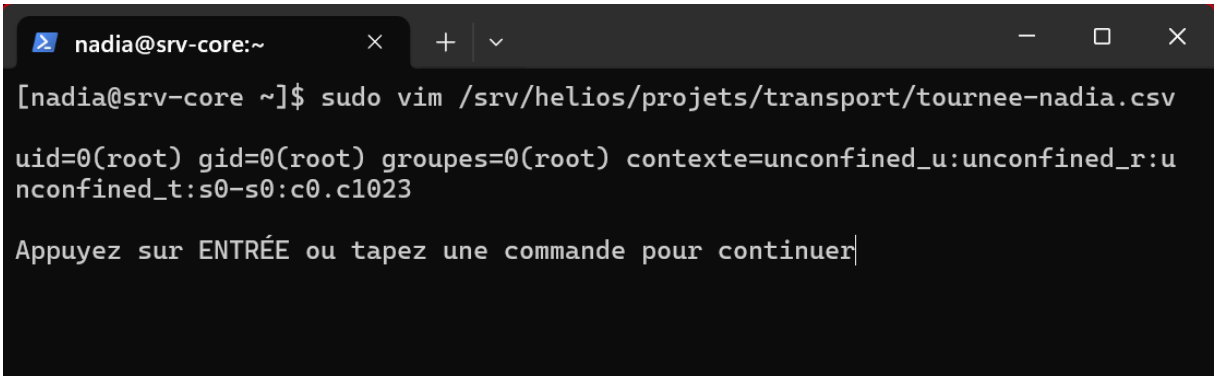


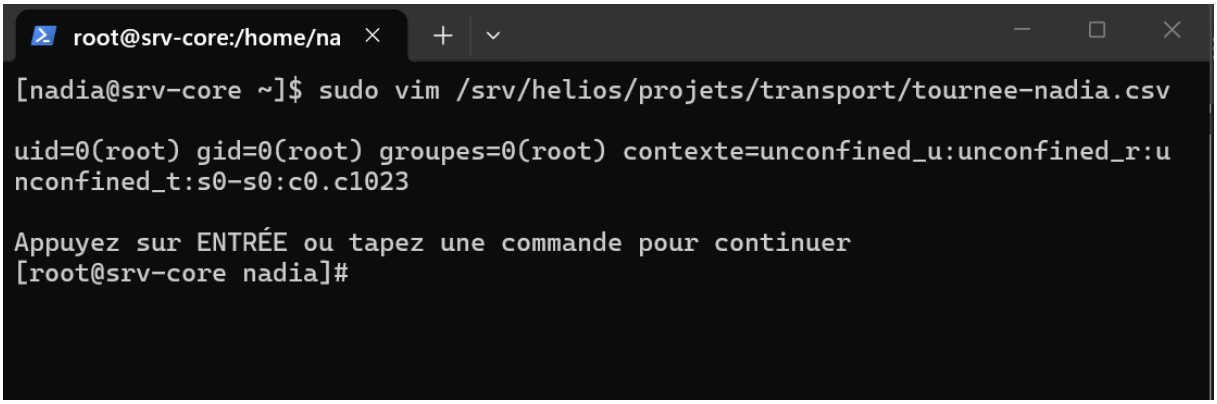
Mission 4.3 :

Ajout mauvaise règle :



Evasion shell :





Autoriser Vim donne un accès presque complet au root.